

Linear algebra: Exercise session 3

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1. In a garden there are only rabbits and chickens. After counting there appear to be 108 legs and 39 heads. How many chickens and rabbits are in the garden?
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2. Describe the set of solutions of the following systems by Gauss-Jordan elimination algorithm:

(a)
$$\begin{cases} 3x_1 + 2x_2 = 5, \\ 6x_1 + 3x_2 = 7, \end{cases}$$

(b)
$$\begin{cases} 3x_1 + 2x_2 + 4x_3 = 5, \\ x_1 + x_2 - 3x_3 = 2, \\ 4x_1 + 3x_2 + x_3 = 7, \end{cases}$$

(c)
$$\begin{cases} x_1 + 2x_2 + x_3 = 0, \\ x_1 - x_2 + 2x_3 = 1, \\ 3x_1 + 5x_3 = 3, \end{cases}$$

(d)
$$\begin{cases} 2x_1 + 3x_2 + x_3 = 1, \\ x_1 + x_2 + x_3 = 3, \\ 3x_1 + 4x_2 + 2x_3 = 4, \end{cases}$$

(e)
$$\begin{cases} x_1 + 2x_2 + x_3 = 11, \\ 2x_1 + x_2 - x_3 = 1, \\ x_1 - x_2 + 3x_3 = 10, \end{cases}$$

(f)
$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 0 & -2 & 2 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 20 \\ 0 \end{bmatrix}.$$

3. Suppose the matrix $A \in \mathbb{R}^{4 \times 4}$ is defined as the product of three squared matrices, i.e.,

$$A = BCD = \begin{bmatrix} 1 & 3 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & -1 & 2 & 1 \end{bmatrix}.$$

Solve $Ax = \begin{bmatrix} 12 & 6 & 32 & 2 \end{bmatrix}^T$ for $x \in \mathbb{R}^4$.

4. Computers are really efficient in doing matrix operations. This is the reason we convert mathematical operations into matrix operations. We are interested to convert operations like *scaling row*, *sum of two rows* and *swap two rows* into multiplication of a squared matrix with augmented matrix, i.e. $P * A$.

Considering augmented matrix A ,

$$A = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right].$$

Find the matrix P for the following operations:

- (a) $r_1 \leftrightarrow r_2$
 - (b) $r_3 \leftrightarrow r_2$
 - (c) $r_1 + r_2 \rightarrow r_1$
 - (d) $cr_1 \rightarrow r_1$
 - (e) $ar_2 + br_3 \rightarrow r_2$
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5. Write a Python program that:

- (a) asks the user to enter a matrix A (the coefficient matrix) and a vector b (the constant terms on the right-hand side of the equations);
 - (b) applies the **Gauss-Jordan elimination algorithm** to solve the linear system $Ax = b$;
 - (c) determines and displays whether the system:
 - has a **unique solution** x^* ,
 - has **infinitely many solutions**, or
 - has **no solution**.
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